# RAJA KODUMURU

AI & Machine Learning Engineer | Generative AI Specialist

8076511746 | rajakodumuru1999@gmail.com | Hyderabad, Telangana, India, 507305

LinkedIn: linkedin.com/in/kodumururaja | GitHub: github.com/KodumuruRaja

## PROFESSIONAL SUMMARY

Results-driven AI & Machine Learning Engineer with 6+ years of experience specializing in Generative AI, Conversational AI, and LLM-powered applications. Proven expertise in building high-performance, low-latency AI systems achieving sub-30ms response times for enterprise applications. Strong track record of delivering scalable solutions across healthcare, financial services, retail, and manufacturing industries using cutting-edge technologies including GPT-4, LLAMA, Mistral, LangChain, and RAG architectures.

## CORE COMPETENCIES

- Generative AI & LLMs (GPT-3.5/4, LLAMA, Mistral)
- LangChain, LangGraph, CrewAI, Llama Index
- RAG Pipelines & Vector Databases
- Conversational AI & Chatbot Development
- Computer Vision (Stable Diffusion, Pix2Pix, OWLViT)
- Cloud Platforms (AWS, Azure, GCP, Vertex AI)
- Python, SQL, Machine Learning
- ETL Pipelines & Data Engineering (Airflow)
- Prompt Engineering & API Integration
- Production Deployment & Model Optimization

## PROFESSIONAL EXPERIENCE

### AI & Machine Learning Engineer

Technonex | Hyderabad, India | April 2026 – Present

- Built a comprehensive Healthcare AI Chatbot capable of generating 80+ distinct medical letters including referral letters, discharge summaries, sick notes, and specialist correspondence, dramatically reducing administrative workload for clinical staff.
- Engineered an intelligent health record summarization system using LLMs and RAG pipelines, enabling clinicians to quickly extract key patient history, diagnoses, and treatment timelines from lengthy unstructured records.
- Developed an automated lab report evaluation module that interprets diagnostic results, flags abnormal values, and provides contextual explanations, improving patient communication and clinical decision support.
- Implemented a medical Q&A engine integrated with domain-specific healthcare knowledge bases, delivering accurate and HIPAA-aware responses to patient and provider queries in real time.
- Leveraged LangChain, GPT-4, and custom prompt engineering to ensure clinical accuracy, consistency, and compliance across all AI-generated healthcare content.

### AI & Machine Learning Engineer

Grid Dynamics | Contractor for Visa Inc. | September 2025 – March 2026

- Architected and deployed 3+ high-performance Generative AI solutions for Visa's global payment infrastructure, achieving sub-30ms latency for real-time transaction processing and fraud detection.
- Engineered production-grade LLM applications using GPT-4, LangChain, and custom RAG architectures to enhance customer service automation, reducing response time by 70% while maintaining 95%+ accuracy.
- Optimized vector database performance (Pinecone, FAISS) for large-scale knowledge retrieval systems, enabling instant access to 10M+ transaction records with millisecond query response times.
- Implemented advanced prompt engineering and fine-tuning strategies (LoRA, QLoRA) for domain-specific financial services use cases, improving fraud pattern detection accuracy by 35%.

### AI Engineer

Gowell Communications (Patternlab.AI) | Hyderabad, India | May 2025 – September 2025

- Led development of industrial IoT conversational bots for real-time monitoring of machine performance, energy consumption, and operational KPIs across manufacturing facilities.
- Built domain-specific knowledge bases using LangChain and Pinecone, enabling accurate Q&A across 100,000+ unstructured documents with 90%+ retrieval precision.
- Designed real-time alerting systems that reduced machine downtime by 25% and improved energy efficiency by 15% through predictive maintenance insights.

### AI & ML Developer (Independent Contractor)

Multiple Client Projects | Hyderabad, India | March 2024 – May 2025

**Healthcare AI Chatbot (US Healthcare Client)**

- Developed HIPAA-compliant conversational AI automating appointment scheduling, medication queries, and 500+ common FAQs with 92% accuracy.

**LinkedIn Jobs Automation Platform**

- Architected an AI-powered job matching system reducing job search time by 60%, processing 1,000+ candidate profiles daily with 85% matching accuracy.

**Retail Customer Service AI Solutions**

- Automated 70% of repetitive customer support queries, increasing conversion rates by 28% and improving customer satisfaction scores by 40%.

**Computer Vision & Image Processing**

- Implemented image processing solutions using Stable Diffusion v1.5, OWL-ViT, and Pix2Pix, processing 10,000+ images daily with 95%+ accuracy.

### AI & ML Engineer

AAPMOR Technologies | Hyderabad, India | January 2024 – February 2024

- Developed enterprise Generative AI apps using LangChain, GPT-4, and Mistral 7B for intelligent conversational workflows across retail and healthcare sectors.
- Engineered RAG-based Q&A systems integrating PDFs, Excel, JSON, and CSV across 50,000+ documents with sub-100ms query latency.

### Software Developer

Tata Consultancy Services (TCS) | Hyderabad, India | February 2022 – January 2024

**Home Depot (Retail Analytics & ML)**

- Designed ML models on GCP for demand forecasting, improving inventory accuracy by 30% and reducing overstock by 22%.

**Nielsen (Media & Data Analytics)**

- Built scalable ETL pipelines using Apache Airflow for 100TB+ datasets, reducing data processing time by 45% with 99.5% data accuracy.

### Data Scientist & ML Engineer

Various Indian Startups | Hyderabad, India | April 2020 – February 2022

- Built end-to-end ML pipelines for ratings analysis and sentiment classification, processing 50,000+ monthly reviews with 88% accuracy.
- Delivered production-ready analytics on GCP in collaboration with Nielsen and Home Depot teams.

### Machine Learning Research Intern

Research Institution | Faridabad, Haryana, India | January 2020 – April 2020

- Designed an ML Automation Framework with drag-and-drop functionality, reducing model development time from hours to seconds across the full ML lifecycle.

## TECHNICAL SKILLS

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| **Languages** | Python, SQL |
| **AI/ML Frameworks** | LangChain, LangGraph, Llama Index, CrewAI, Hugging Face, Scikit-learn |
| **LLMs** | GPT-3.5/4, LLAMA, Mistral, Amazon Bedrock, Claude |
| **Vector Databases** | Pinecone, FAISS, ChromaDB |
| **Databases** | MySQL, PostgreSQL, MongoDB |
| **Cloud Platforms** | AWS, Microsoft Azure, Google Cloud Platform (GCP), Google Vertex AI |
| **Computer Vision** | Stable Diffusion, Pix2Pix, OWL-ViT |
| **Data Engineering** | Apache Airflow, ETL Pipelines, Pandas, NumPy |

**AI Specializations** Conversational AI, RAG Pipelines, Prompt Engineering, Fine-tuning (LoRA, QLoRA, Full Fine-tuning), Model Optimization

## EDUCATION

**Bachelor of Technology (B.Tech) in Computer Science Engineering**
Faridabad, Haryana, India

## KEY ACHIEVEMENTS

- Achieved sub-30ms latency across 3 production Generative AI systems at Visa, processing millions of real-time transactions daily.
- Reduced customer response times by 70% through implementation of advanced RAG-based conversational AI systems.
- Improved inventory forecasting accuracy by 30% using GCP-based ML models for retail analytics at Home Depot.
- Built HIPAA-compliant healthcare chatbot with 92% accuracy in patient query responses.
- Automated 70% of customer support queries for retail clients, increasing conversion rates by 28%.
- Reduced data processing time by 45% through optimized ETL pipeline development at Nielsen.
- Developed Healthcare AI system at Technonex generating 80+ medical letter types with end-to-end health record summarization and lab report evaluation.